

ALEX ANKOMAH

Telecom Network Planning & Optimization Engineer (RAN/RNO/NPO)

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PROFESSIONAL SUMMARY

Telecom Network Planning & Optimization Engineer with 20+ years of experience in RAN operations, integration, and optimization across multi-vendor environments — Huawei, Ericsson, Nokia, and Alcatel-Lucent. Proven expertise spanning 2G, 3G, 4G LTE, and 5G NSA mobile technologies, including gNodeB/eNodeB/NodeB/BTS integration, parameter tuning, and KPI performance improvement, across deployments in Ghana, Burundi, Niger, Senegal, and Guinea-Bissau. Currently supporting national-level network optimization and regulatory compliance for ITS/ARCT, the Burundi National Telecom Regulator. Hands-on professional skilled in OSS configuration (U2020, ENM, NetAct, OSS-RC), drive testing, spectrum analysis, and inter-RAT optimization, with a strong record of award-winning performance and technical leadership.

CORE COMPETENCIES

- RAN Operations, Integration & Optimization (2G/3G/4G LTE/5G NSA)
- Network Planning & Optimization (NPO) / Radio Network Optimization (RNO)
- Multi-Vendor RAN: Huawei, Ericsson, Nokia, Alcatel-Lucent, ZTE
- Node Integration: gNodeB, eNodeB, NodeB, BTS, RNC, BSC
- OSS & Node Management: U2020, ENM, NetAct, OSS-RC, WebLMT, Moshell
- Drive Testing & Post-Processing: TEMS, Nemo, Actix
- Radio Planning & Auditing: Atoll, MapInfo, Google Earth
- Spectrum Analysis & Interference Resolution: Anritsu Spectrum Analyzer
- Feature Activation: VoLTE, CSFB, Carrier Aggregation, MIMO, DC-HSPA, NSA
- KPI Analysis, QoS Monitoring & Regulatory Compliance
- Reporting & BI: Power BI, Tableau, QlikView, Business Objects, Excel
- Database & SQL: Oracle, MySQL, Sybase
- Project Management & Technical Team Leadership
- Languages: English (fluent), French (intermediate)

KEY ACHIEVEMENTS

- Contributed as part of the engineering team that designed and deployed UMTS network technology across Sub-Saharan Africa.
- Zain Achievers Innovation Award (2010) for developing an SMS-based application enabling sales teams to track cell traffic and availability nationwide.
- Key member of the 3G U900 refarming project expanding rural network coverage in Senegal (2015).
- Recognized as a top-performing engineer during the Huawei-led AirtelTigo Network Swap project.
- Huawei Performance Award recipient (cash prize and monthly achiever award) for outstanding technical contribution to network improvement.
- Deployed 42 Mbps DC-HSPA dual-carrier activation across 81 sites for Orange Niger, improving network capacity.

PROFESSIONAL EXPERIENCE

Network Planning & Optimization Engineer (NPO)

Feb 2023 – Present

ITS / ARCT Burundi — National Telecom Regulatory Authority — Bujumbura, Burundi

- Monitor and analyze key QoS network performance parameters — call completion rate, drop rate, throughput, and voice quality — across three mobile network operators to ensure regulatory compliance.
- Audit Huawei and Nokia network configuration dumps, radio/logical/physical parameters, and feature settings against network statistics to identify congestion and low-throughput areas.

- Conduct network planning analysis using Atoll for coverage prediction, capacity validation, and quality assessment against regulatory standards.
- Lead quarterly national drive testing campaigns to benchmark operator performance against regulatory KPIs.
- Provide technical support resolving interference issues, including cross-border interference with DR Congo, Rwanda, and Tanzania.
- Conduct quarterly spectrum scanning to verify operator compliance with allocated frequency and bandwidth across GSM, UMTS, and LTE.

Drive Test & Optimization Engineer

May 2022 – Feb 2023

Emerald Telecom — Huawei / MTN Ghana Project — Accra, Ghana

- Processed customer complaint tickets and performed drive testing with Huawei PHU, Nemo Handy, and TEMS to diagnose coverage, quality, and throughput issues.
- Executed physical antenna optimization changes on-site and validated improvements via drive testing across 2G/3G/4G.
- Conducted Single Site Verification (SSV) and cluster acceptance testing, validating PCI/PSC/BCCH, antenna height, azimuth, LAC, TAC, and Cell ID against planned data.
- Performed feature activation testing for MIMO, VoLTE, CSFB, and Carrier Aggregation.

Radio Network Optimization Engineer (RNO)

Sept 2021 – Feb 2022

Emerald Telecom — Accra, Ghana

- Identified high-utilization 4G cells (PRB, power, CCE) and recommended capacity optimization solutions.
- Monitored KPIs — call setup success rate, drop rate, throughput — and proposed parameter and feature changes to MTN.
- Identified 3G/4G coverage overlaps, overshooting cells, poor SINR/RSRP/RSRQ, and low-throughput areas, recommending corrective changes.
- Used Anritsu Spectrum Analyzer to investigate and resolve external interference sources.

3G RNO Consultant

Jan 2016 – Mar 2016

Huawei — Orange Guinea-Bissau — Bissau, Guinea-Bissau

- Led optimization of an 80-site 3G plus 18-site WiMax (Alvarion) managed services project.
- Performed parameter tuning, neighbor optimization, and feature activation on RNC/NodeB to improve QoS.
- Conducted capacity analysis of baseband boards and activated software licenses.

3G RNO Engineer

Apr 2015 – Jan 2016

Alcatel-Lucent / Nokia — Orange Senegal — Dakar, Senegal

- Performed pre-integration checks and neighbor optimization for U900/U2100 sites.
- Activated Dual-Cell DC-HSPA and 3rd carrier features to improve data throughput and relieve congestion.
- Analyzed weekly worst-performing cells and proposed remediation plans.

3G RNO Engineer

Nov 2014 – Feb 2015

Alcatel-Lucent — Orange Niger — Niamey, Niger

- Activated 42 Mbps DC-HSPA throughput (dual-frequency F2+F3) across 81 sites.
- Added 3rd carrier to 81 sites to relieve CE, power, and DL code congestion.
- Integrated IUR interfaces and split 81 NodeB sites between two RNCs.

Manager, Network Performance & Optimization (NPO)

Apr 2011 – Nov 2014

Glo Mobile Ghana — Accra, Ghana

- Led pre-tuning and verification for a greenfield 3G rollout project.
- Managed RNC license activation, NodeB CE configuration, and software upgrades.
- Directed neighbor audits and daily QoS monitoring across GSM and UMTS networks.

RF Planning & Optimization Engineer

Nov 2010 – Mar 2011

Airtel Ghana — Accra, Ghana

- Conducted weekly radio parameter consistency checks using Ericsson Explorer.
- Led software upgrade impact analysis for OSS-RC (5.3 → 10.3) and RNC/RBS (P6 → P7).

- Built BI dashboards in ENIQ/Business Objects for KPI and capacity trend analysis.

RF Performance & Optimization Engineer

Sept 2008 – Mar 2010

Zain Ghana — Accra, Ghana

- Analyzed and resolved top-20 worst-performing cells network-wide.
- Monitored CS/PS RAB and HSDPA/HSUPA setup success rates, identifying throughput bottlenecks.
- Managed CE software/hardware upgrade requests and SLA-driven KPI/counter governance.

Assistant Telecom Manager / Principal Telecom Officer

Oct 2001 – Aug 2008

Ghana Telecom — Accra & Koforidua, Ghana

- Monitored core and radio network QoS, traffic congestion, and blocking across MSC, PSTN, and BSC/BTS.
- Managed PSTN interconnect routes, subscriber line provisioning, and monthly traffic/billing reporting.

EDUCATION

BSc, Information Technology — Methodist University College of Ghana, Accra, Ghana (2008)

Associate Degree, Telecommunication Engineering — Multimedia University, Melaka, Malaysia (2001)

CERTIFICATIONS & PROFESSIONAL TRAINING

- Project Management — Glenridge Consulting Group LLC, USA (2009)
- Data Science & Analytics — IBM Digital Nation Africa
- Tableau Professional — Simplilearn (2022)
- Ericsson, Huawei & Astellia GSM/UMTS/LTE Network Optimization Training (2006–2010)